

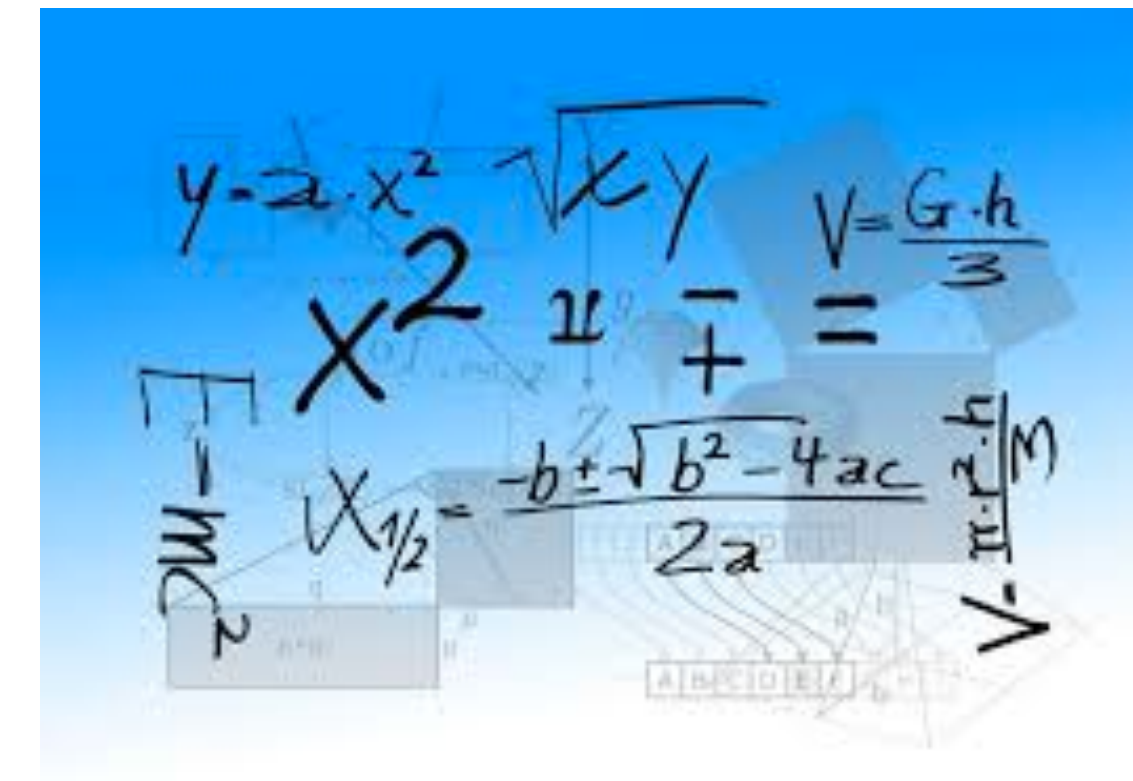
HAYDE MARTINEZ - DE CERO A CIENCIA DE DATOS

AN OPEN DOOR TO THE WORLD OF AI

Let's talk about AI!

@haydemtzi

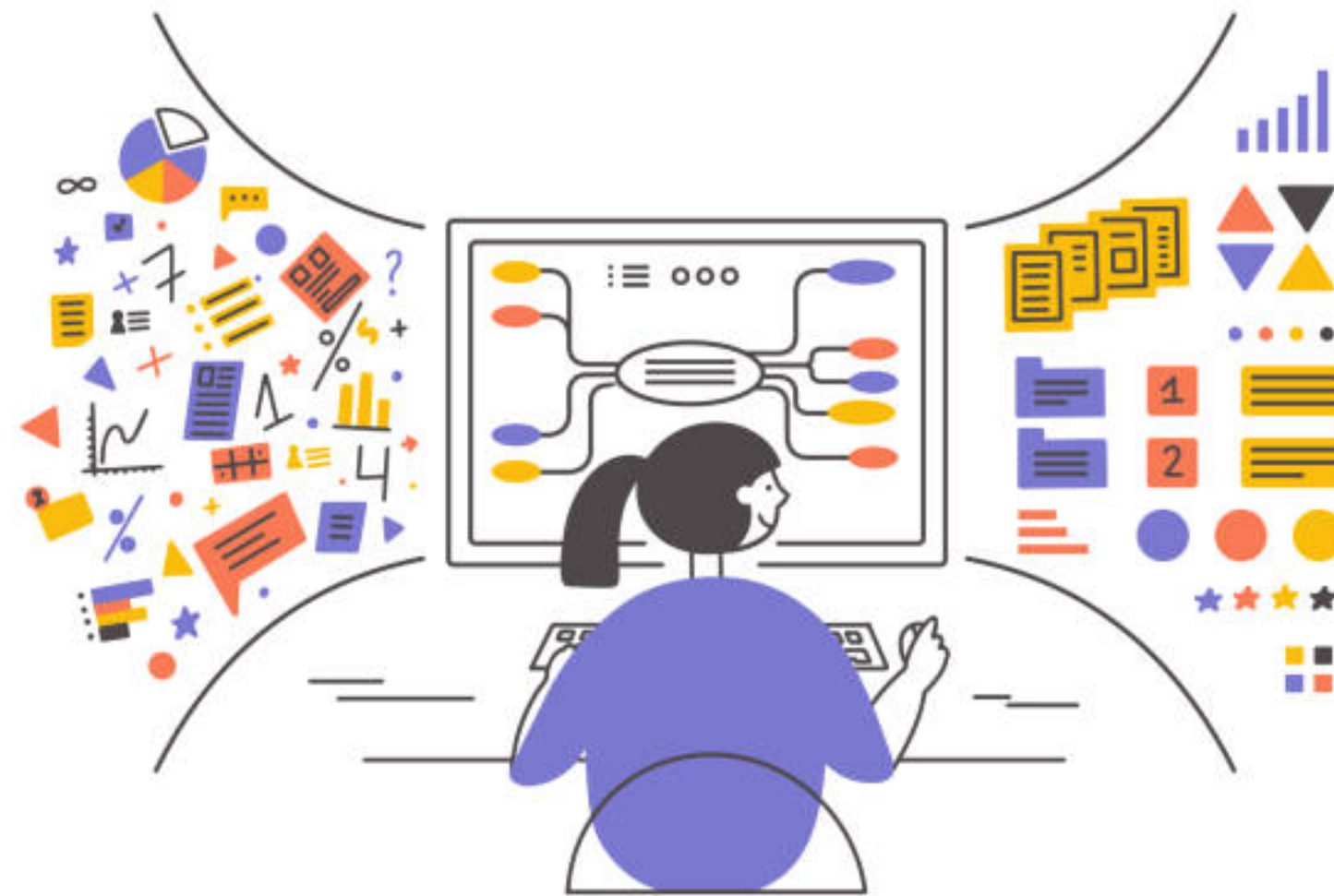
MODEL VS ALGORITHM



Algorithm	A set of instructions to solve a task or a problem	An empty tupper -> the structure is there, but it needs data to become useful	It's the logic used to train or apply a model (e.g. Linear regression)
Model	The result of applying an algorithm to data -> a learned system	A tupper with food -> ready to be used, already has knowledge inside	It's what actually makes predictions, classifications, or generates outputs.

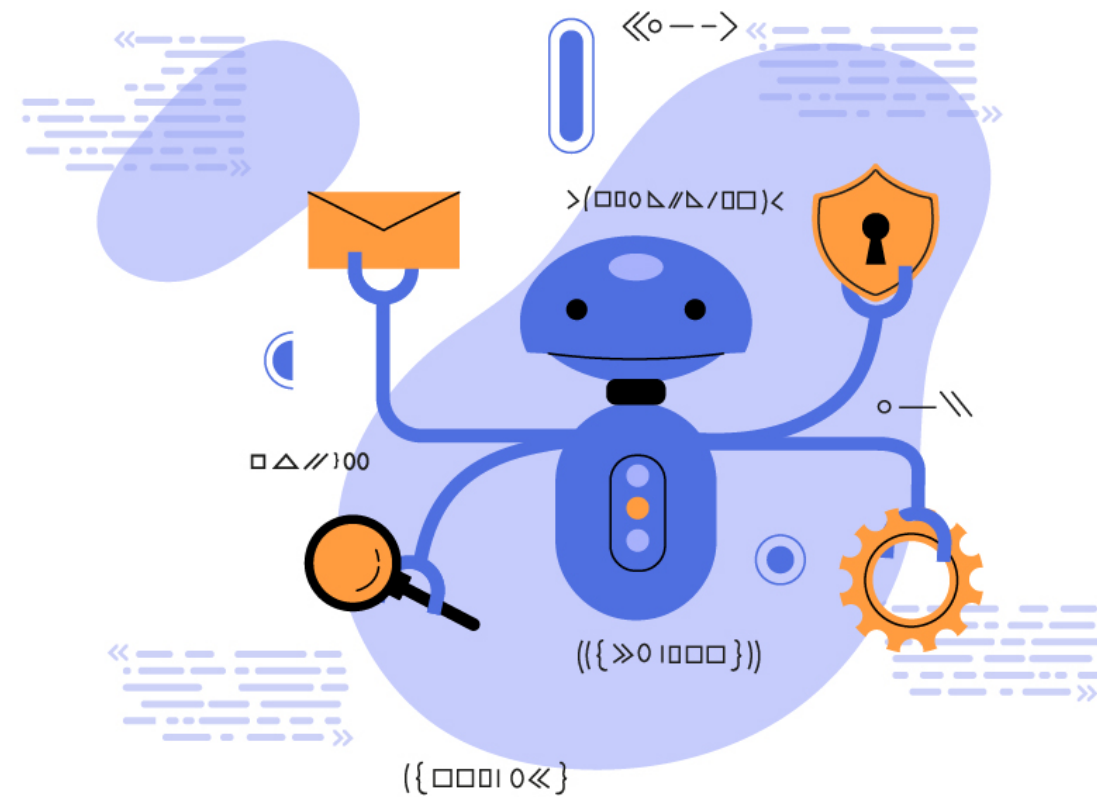
DATA SCIENCE

- **By definition:**
Data science is an interdisciplinary field that involves scientific methods, processes, and systems to extract knowledge or gain deeper understanding from data in its various forms, whether structured or unstructured. It builds upon fields like statistics, data mining, machine learning, and predictive analytics.



MACHINE LEARNING

- **By definition:**
Is a technique focused on developing algorithms and models that enable computers to learn from huge amounts of data and make predictions or decisions without being explicitly programmed for each task.



ARTIFICIAL INTELLIGENCE

- **By definition:**
Is a field of computer science focused on creating systems or machines capable of performing tasks that typically require human intelligence. These tasks include learning, reasoning, problem-solving, perception, and language understanding



Data Science is a broad field that combines statistical analysis, data visualization, and predictive modeling. It involves exploring and understanding data in order to extract insights and build models that help answer questions and solve real-world problems. 📊

Machine Learning (ML) is a subset of Data Science focused on creating models that automatically learn from data to make predictions or classifications without being explicitly programmed for each task. 🤖🔍

Artificial Intelligence (AI) is the overarching domain that includes ML and other techniques aimed at building systems capable of intelligent behavior such as reasoning, learning, problem-solving, or mimicking aspects of human cognition. 🌐🚀

Data Science is like a laboratory of ideas

Machine Learning is the set of automatic tools we develop

Artificial Intelligence is when those tools make decisions and act on their own.



Example:

Exploring transaction patterns and visualizing unusual behaviors → **Data Science**

A model that detects potential banking fraud based on those patterns → **Machine Learning**

A system that automatically blocks suspicious transactions without human intervention → **Artificial Intelligence**



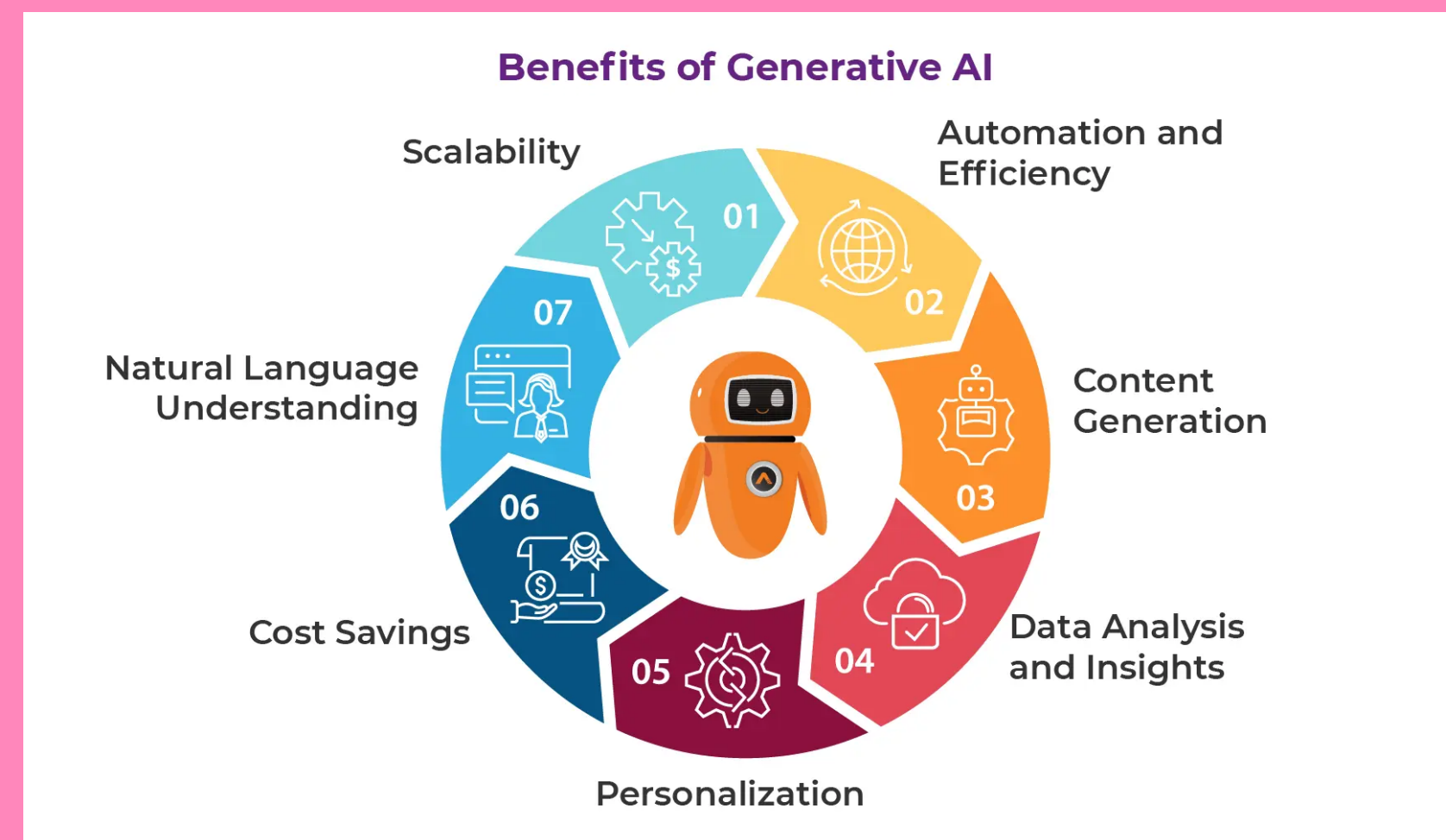
Analysis, cleansing, visualization, simple models	Data Science
Predictive model trained to classify or predict	Machine Learning
Autonomous system that makes decisions using ML and acts	Artificial Intelligence

GENERATIVE AI

- **By definition:**

Is a type of **artificial intelligence** focused on **creating new content** (like text, images, audio, or code) based on patterns **learned from large datasets**.
It doesn't just analyze information; it generates original outputs.

When we interact with tools like **ChatGPT, Claude, or DALL·E**, we're using pre-trained models that rely on Data Science (to understand the data) and Machine Learning (to learn from it). ✨🤖



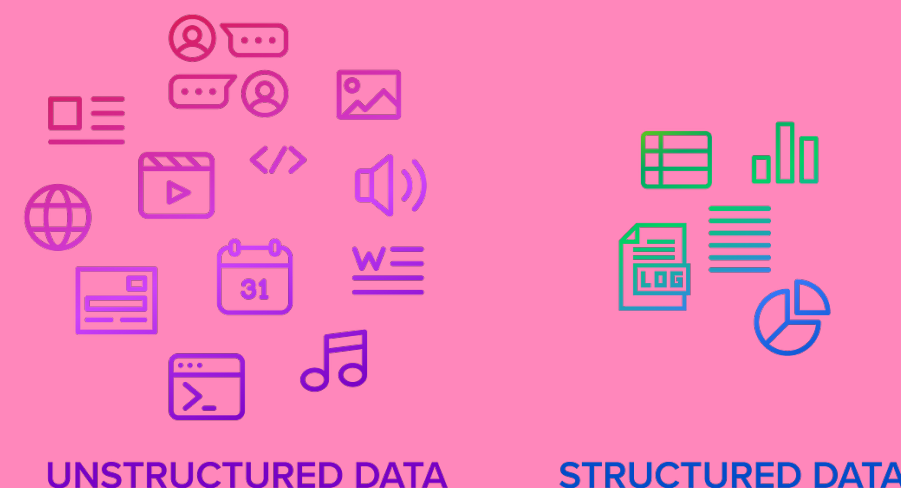
RAG (RETRIEVAL-AUGMENTED GENERATION)

Technique that **combines retrieval and generation** to improve accuracy and relevance.

- **1 Retrieves** relevant information from external sources (like documents, PDFs, or knowledge bases) using semantic search or vector databases.
- **2 Generates** a response using that information as context, rather than relying only on pre-trained knowledge.

Ideal for working with **unstructured data** and answering domain-specific questions.

Used in **chatbots, enterprise assistants**, and internal Q&A tools to deliver **grounded, real-time responses**.



💡 Example:

A chatbot trained on general knowledge → **Generative AI**

A chatbot that also searches your company's internal documents before answering → **RAG**

AGENTS

AI systems designed to **take actions toward a goal**.

They can **make decisions, use tools**, and **interact with environments** following a sequence of steps, not just generating a single response.

Unlike a regular chatbot that just replies to a question, an agent can **break down a task, search for information, use external tools**, and **adapt based on the outcome**.

💡 **Example:**

A chatbot that gives you a one-time answer → **Generative AI**

A system that books your flight, checks your calendar, and compares options step by step → **Agent**



AGENTIC AI

Is an emerging paradigm where AI systems act with **purpose, autonomy, and memory**.

Instead of waiting for one prompt at a time, these systems can **plan, remember past interactions, and choose the best strategy** to solve a problem, like a human assistant that learns over time.

Agentic AI combines **Generative AI + decision-making + tool use + memory**, moving us closer to truly intelligent behavior.



💡 Example:

An agent that performs a multi-step task in one session → **Agent**

A system that learns your preferences over time, adapts strategies, and works on goals across sessions → **Agentic AI**

MCP (MODEL CONTEXT PROTOCOL)

Is a way of **structuring the interaction** between a large language model and its environment.

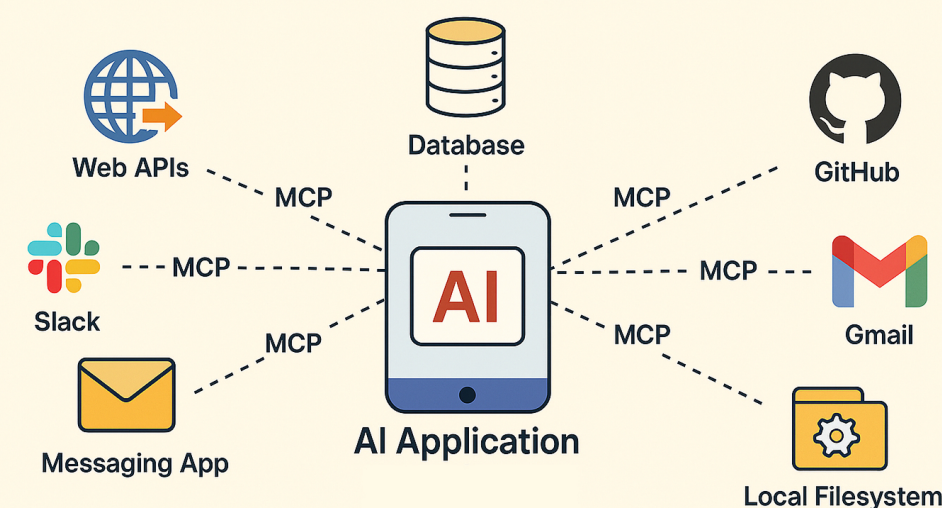
It defines the model's **role, input sources, tools, memory**, and **behavioral rules** so it can respond in a more **reliable, controlled, and context-aware** way.

Instead of just sending an open-ended prompt, **MCP establishes a protocol**:

“This is who you are, this is the information you can access, these are the tools you can use, and this is how you should respond.”

MCP can orchestrate complex interactions, whether the model pulls context from **structured databases, unstructured documents**, or **external APIs** making it ideal for **enterprise-grade, multi-turn, and goal-driven applications**.

It's like setting up the **stage, role, script, and tools** before the AI starts performing.



💡 Example:

Sending a plain prompt to a model → **Generative AI**

Designing a structured context where the model has a role, memory, data access, and tool use → **MCP**

COOKING WITH INTELLIGENCE (AND STRUCTURE)

Imagine you're preparing a meal at home. Everything you do — your choices, your setup, your knowledge — represents a concept:

Agent = You following a recipe

You know what dish you're making and follow the steps: chop, season, cook. You make small decisions (a bit more salt?), but overall, you're sticking to a defined process.

Agentic AI = You cooking without a recipe, remembering who you're cooking for

You know your partner doesn't like garlic, and it's a cold day, so you want to make something cozy. You adjust the flavor, taste as you go, and recall what's worked before. You're **making thoughtful, personalized decisions** to reach a goal — not just following instructions.

RAG (Retrieval-Augmented Generation) = Your recipe book

Before starting, you check your favorite cookbook or handwritten notes. You don't invent everything from scratch — you **retrieve reliable information first**, and then decide how to proceed.

MCP (Model Context Protocol) = Your full kitchen setup

Before you cook, you define everything you need:

What ingredients do you have?

What tools can you use?

Who are you cooking for?

Are there any rules? (no spice, vegetarian, etc.)

MCP is the **structure, context, and constraints** that guide the process — you're not improvising randomly, you have **intention, tools, and a plan**.

	What it is	Main Use	When to use it alone	When to combine
RAG	A method that retrieves external information before generating a response	Grounding model outputs in up-to-date or proprietary knowledge	When you only need factual, document-based answers (e.g., internal search bots, Q&A over PDFs)	Use it inside Agentic AI as a tool for decision-making
Agentic AI	AI that acts with autonomy, memory, and planning	Solving multi-step tasks through reasoning, tool use, and interaction	When your goal is autonomous execution (e.g., research agents, automation flows)	Combine with RAG for reliable knowledge + MCP to orchestrate behavior
MCP	A structured interface to control how an AI model behaves (roles, tools, memory, instructions)	Defining how the model interacts and what it can do	When you need control, consistency, or modularity in GenAI apps (e.g., assistants with clear behavior)	Use it as the foundation for Agentic AI systems or complex RAG pipelines



WHO BUILDS AI?

Data Engineer

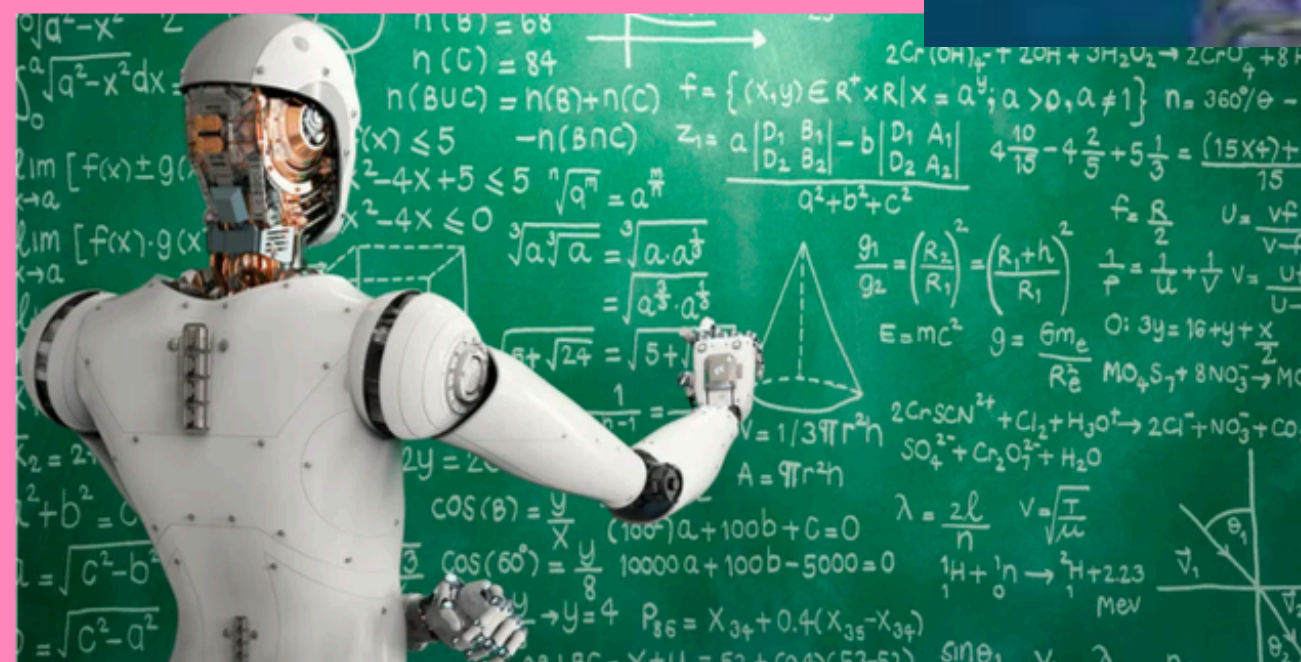


Machine Learning Engineer



AI Engineer

Data Scientist



Data Analyst





THANK YOU!

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